2.60 Moments

Worked Solutions

1 a)

M(O) $R(\circlearrowleft)$: $8 \times 3 = 24$ Nm anti-clockwise

1 b)

M(O) $R(\circlearrowleft)$: $-10 \times 5 = -50$ Nm

50 Nm clockwise

1 c)

M(O) $R(\circlearrowleft)$: $1.6 \times 2 = 3.2$ Nm anti-clockwise

1 d)

M(O) $R(\circlearrowleft)$: $-20 \times 2 = -40$ Nm

40 Nm clockwise

1 e)

M(O) $R(\circlearrowleft)$: $-4 \times 3 - 6 \times 2 = -24$ Nm

24 Nm clockwise

1 f)

M(O) $R(\circlearrowleft)$: $9 \times 0.5 - 7 \times 1.2 = -3.9$ Nm

3.9 Nm clockwise

2 a)

M(A) $R(\circlearrowleft)$: $5 \times 2 = 10$ Nm anti-clockwise

2 b)

M(B) $R(\circlearrowleft)$: $5 \times 2 = 10$ Nm anti-clockwise

2 c)

M(C) $R(\circlearrowleft)$: $5 \times 0 = 0$ Nm

2 d)

M(D) $R(\circlearrowleft)$: $5 \times 0 = 0$ Nm

2 e)

M(O) $R(\circlearrowleft)$: $5 \times 1 = 5$ Nm anti-clockwise

3 a)

M(A) $R(\circlearrowleft)$: $12 \times 0 = 0$ Nm

3 b)

M(B) $R(\circlearrowleft)$: $12 \times 0 = 0$ Nm

3 c)

M(C) $R(\circlearrowleft)$: $-12 \times 4 = -48$ Nm

48 Nm clockwise

3 d)

M(D) $R(\circlearrowleft)$: $-12 \times 4 = -48$ Nm

48 Nm clockwise

3 e)

M(O) $R(\circlearrowleft)$: $-12 \times 2 = -24$ Nm anti-clockwise

24 Nm clockwise

4 a)

M(A) $R(\circlearrowleft)$: $15 \times 0 - 10 \times 5 = -50$ Nm

50 Nm clockwise

4 b)

M(B) $R(\circlearrowleft)$: $15 \times 0 + 10 \times 0 = 0$ Nm

4 c)

M(C) $R(\circlearrowleft)$: $-15 \times 6 + 10 \times 0 = -90$ Nm

90 Nm clockwise

4 d)

M(D) $R(\circlearrowleft)$: $-15 \times 6 - 10 \times 5 = -140$ Nm

140 Nm clockwise

4 e)

M(O) $R(\circlearrowleft)$: $-15 \times 3 - 10 \times 2.5 = -70$ Nm

70 Nm clockwise

5 a)

$R(\rightarrow)$: $10 - X = 0 \therefore X = 10$

5 b)

$R(\uparrow)$: $Y - 20 = 0 \therefore Y = 20$

5 c)

M(A) $R(\circlearrowleft)$: $0.04 \times 20 - 10d = 0 \therefore d = 0.08$ m

5 d)

Let k be the distance that the force has moved from B to C

M(A) $R(\circlearrowleft)$:

$0.04 \times 20 - 0.08 \times 10 + 10k$

$= 0 + 10k$

k increases from 0 to 0.2 which creates a resultant moment. Therefore, the forces have a resultant moment which have an anti-clockwise turning effect on the lamina

6 a)

M(P) $R(\circlearrowleft)$: $6 \times 3 - 4 \times 6 + 11 \times 3 = 27$ Nm

27 Nm anti-clockwise

6 b)

M(A) $R(\circlearrowleft)$: $-4 \times 3 + 3 \times 8 = 12$ Nm

12 Nm anti-clockwise

6 c)

M(B) $R(\circlearrowleft)$: $3 \times 5 - 6 \times 3 = -3$ Nm

3 Nm clockwise

6 d)

M(C) $R(\circlearrowleft)$: $-6 \times 8 + 4 \times 5 = -28$ Nm

28 Nm clockwise

6 e)

M(Q) $R(\circlearrowleft)$: $-6 \times 13 + 10 \times 4 - 3 \times 5 = -53$ Nm

53 Nm clockwise

7 a)

$R(\uparrow)$: $P + Q = 50$

M(P) $R(\circlearrowleft)$: $-50 \times 2.5 + 5Q = 0 \therefore Q = 25$ N

$P = 50 - 25 = 25$ N

7 b)

$R(\uparrow)$: $P + Q = 110$

M(A) $R(\circlearrowleft)$: $-110 \times 4 + 11Q = 0 \therefore Q = 40$ N

$P = 110 - 40 = 70$ N

8 a)

$R(\uparrow)$: $P + Q = 30$

M(P) $R(\circlearrowleft)$: $-30 \times 5 + 8Q = 0 \therefore Q = 18.75$ N

$P = 30 - 18.75 = 11.25$ N

8 b)

$R(\uparrow)$: $P + Q = 69$

$\mathrm{M}(P)$ $R(\circlearrowleft)$: $-5 \times 4 - 64 \times 7 + 13Q = 0$

$\therefore Q = 36$ N

$P = 69 - 36 = 33$ N

8 c)

$R(\uparrow)$: $P + Q = 5$

$\mathrm{M}(P)$ $R(\circlearrowleft)$: $0.65Q - 5 \times 0.25 = 0$

$\therefore Q = \frac{25}{13} = 1.92$ N (3 SF)

$P = 5 - \frac{25}{13} = \frac{40}{13} = 3.08$ N (3 SF)

9 a)

$R(\uparrow)$: $15 + 25 = W \therefore W = 40$

$\mathrm{M}(A)$ $R(\circlearrowleft)$: $-40x + 25 \times 0.5 = 0$

$\therefore x = 0.3125$

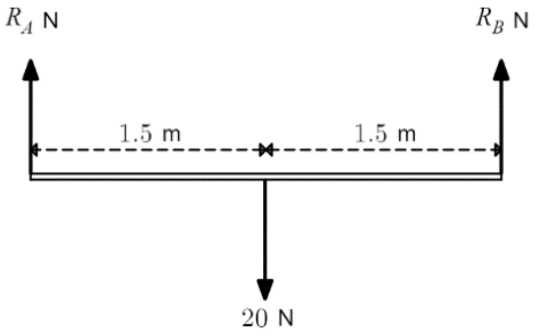
9 b)

$R(\uparrow)$: $P + Q = 120$

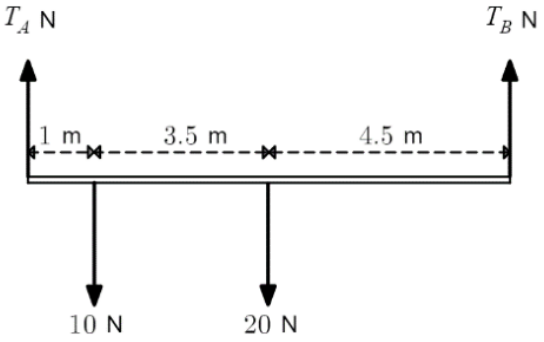
$\mathrm{M}(A)$ $R(\circlearrowleft)$: $-120 \times 0.3 + 0.5Q = 0 \Rightarrow Q = 72$

$\therefore P = 120 - 72 = 48$

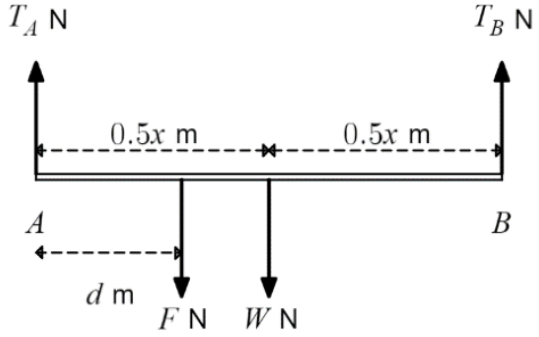
10 a)



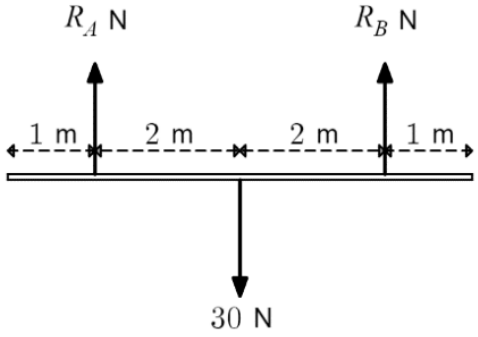
10 b)



10 c)



11)

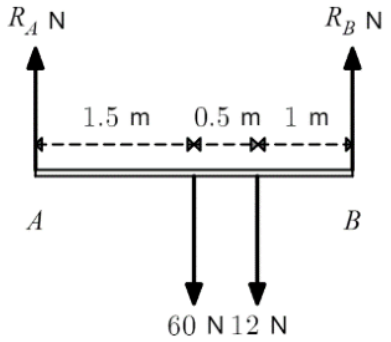


$R(\uparrow)$: $R_A + R_B = 30$

M(A) $R(\circlearrowleft)$: $-30 \times 2 + 4R_B = 0 \Rightarrow R_B = 15$

$\therefore R_A = 30 - 15 = 15$

12)



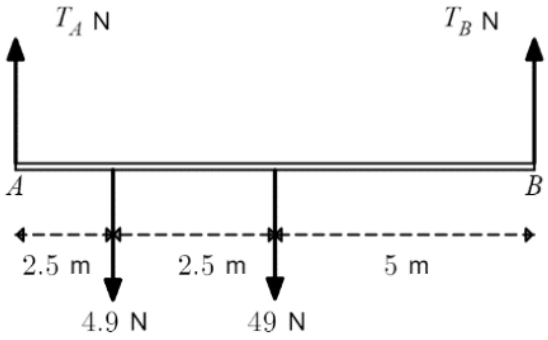
$R(\uparrow)$: $R_A + R_B = 62$

M(A) $R(\circlearrowleft)$: $-60 \times 1.5 - 12 \times 2 + 3R_B = 0$

$\Rightarrow R_B = 38$ N

$\therefore R_A = 72 - 38 = 34$ N

13)



$R(\uparrow)$: $T_A + T_B = 53.9$

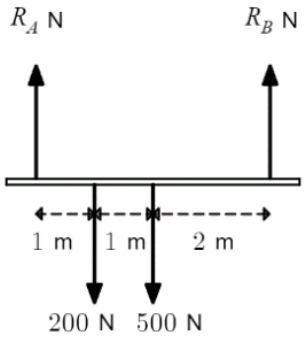
M(A) $R(\circlearrowleft)$: $-4.9 \times 2.5 - 49 \times 5 + 10T_B = 0$

$\Rightarrow T_B = 25.725$ N, $T_A = 53.9 - 25.725 = 28.175$ N

$\therefore T_B = 25.7$ N, $T_A = 28.2$ N (3 SF)

$T_B = 26$ N, $T_A = 28$ N (2 SF – AQA)

14 a)



$R(\uparrow)$: $R_A + R_B = 700$

M(A) $R(\circlearrowleft)$: $-200 \times 1 - 500 \times 2 + 4R_B = 0$

$\Rightarrow R_B = 300$ N

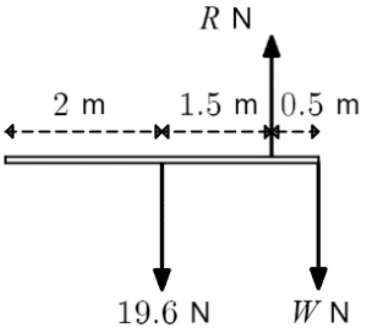
$\therefore R_A = 700 - 300 = 400$ N

14 b)

The plank is modelled as a uniform rod.

The object is modelled as a particle.

15 a)



$R(\uparrow)$: $R = W + 19.6$

M(B) $R(\circlearrowleft)$: $19.6 \times 2 - R \times 0.5 = 0$

$\Rightarrow R = 78.4$ N

15 b)

$\therefore W = 78.4 - 19.6 = 58.8$ N

$W = 59$ N (2 SF – AQA)

15 c)

$m = \frac{W}{9.8} = \frac{58.8}{9.8} = 6$ kg

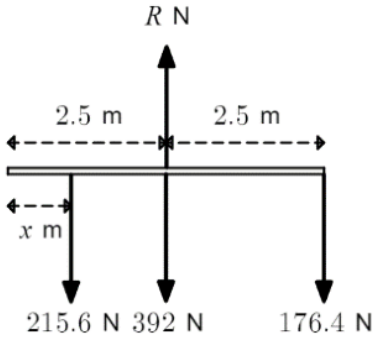
15 d)

$R(\uparrow)$: $R = W + W + 19.6$

$\Rightarrow R = 58.8 + 58.8 + 19.6 = 137.2$ N

$R = 140$ N (2 SF – AQA)

16)



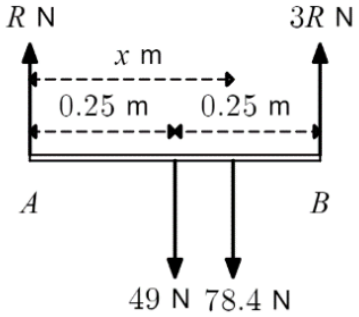
$R(\uparrow)$: $R = 215.6 + 392 + 176.4 = 784$

M(A) $R(\circlearrowleft)$: $-215.6x - 392 \times 2.5 + 2.5 \times 784 - 5 \times 176.4 = 0$

$x = \frac{5}{11} = 0.455$ m (3 SF)

$x = \frac{5}{11} = 0.50$ m (2 SF – AQA)

17)



Let $R_A = R$ so $R_B = 3R$

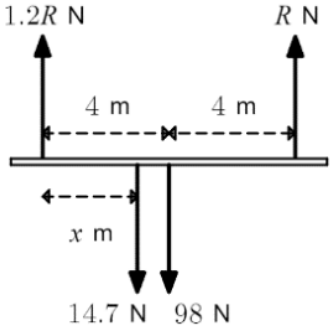
$R(\uparrow)$: $R + 3R = 127.4 \therefore R = 31.85$

M(A) $R(\circlearrowleft)$: $-78.4x - 49 \times 0.25 + 0.5 \times 95.55 = 0$

$x = \frac{29}{64} = 0.453$ m (3 SF)

$x = 0.45$ m (2 SF – AQA)

18)



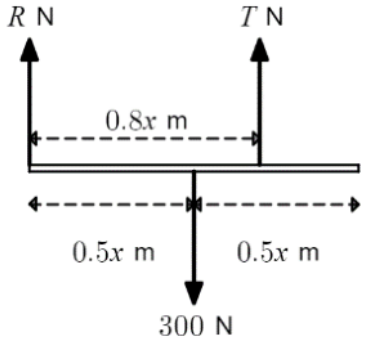
$R(\uparrow)$: $1.2R + R = 112.7 \therefore R = \frac{1127}{22}$

M(A) $R(\circlearrowleft)$: $-14.7x - 98 \times 4 + 8 \times \frac{1127}{22} = 0$

$x = \frac{40}{33} = 1.21$ m (3 SF)

$x = 1.2$ m (2 SF – AQA)

19)



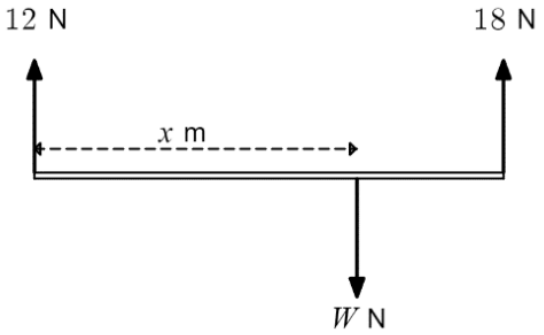
$R(\uparrow)$: $R + T = 300$

M(A) $R(\circlearrowleft)$: $-300(0.5x) + T(0.8x) = 0$

$\Rightarrow T = 187.5$ N

$R = 300 - 187.5 = 112.5$ N

20)

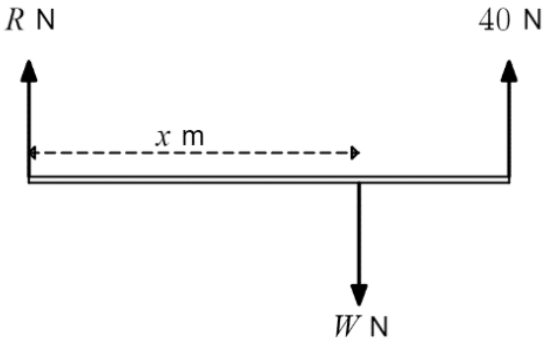


$R(\uparrow)$: $12 + 18 = W \therefore W = 30$

$\mathrm{M}(A)$ $R(\circlearrowleft)$: $-30x + 3 \times 18 = 0$

$\Rightarrow x = 1.8$ m

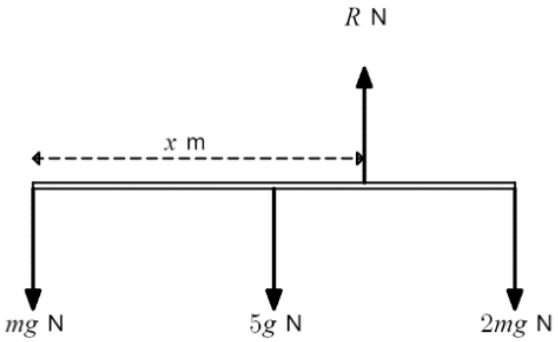
21)



$\mathrm{M}(A)$ $R(\circlearrowleft)$: $-Wx + 10 \times 40 = 0$

$\Rightarrow x = \frac{400}{W}$ m

22)

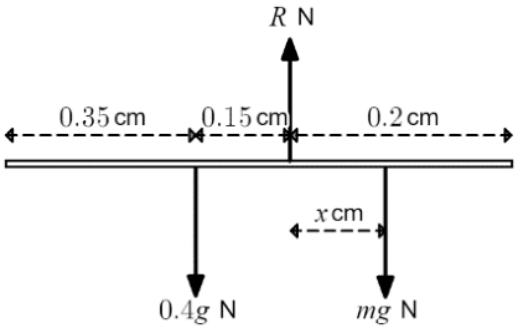


$R(\uparrow)$: $R = mg + 2mg + 5g \therefore R = (3m + 5)g$

$\mathrm{M}(A)$ $R(\circlearrowleft)$: $(3m + 5)gx - 5g \times 2.5 - 2mg \times 5 = 0$

$x = \frac{10mg+12.5g}{(3m+5)g} = \frac{10m+12.5}{3m+5} = \frac{20m+25}{6m+10}$

23 a)



$R(\uparrow)$: $R = (0.4 + m)g$

M(C) $R(\circlearrowleft)$: $0.4g \times 0.15 - mgx = 0$

$\Rightarrow mx = 0.06 \Rightarrow mx = \frac{0.06}{m} = \frac{3}{50m}$ as req.

23 b)

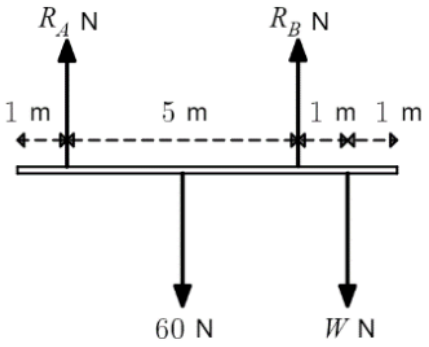
$m = \frac{3}{50x}$

For maximum m, $x = 0.2$

$m = \frac{3}{50 \times 0.2} = 0.3$ kg

So $0 < m \leq 0.3$

24)

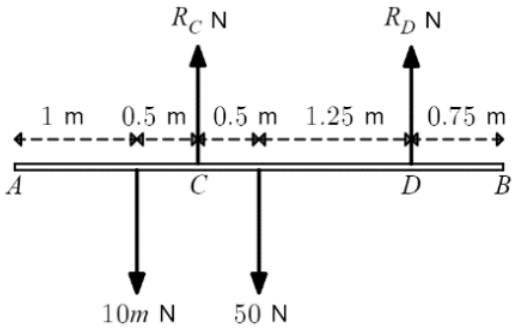


Point of tilting: $R_A = 0$ so $R(\uparrow)$: $R_B = 60 + W$

M(B) $R(\circlearrowleft)$: $60 \times 2 - W \times 1 = 0 \Rightarrow W = 120$ N

$\therefore R_B = 60 + 120 = 180$ N

25)



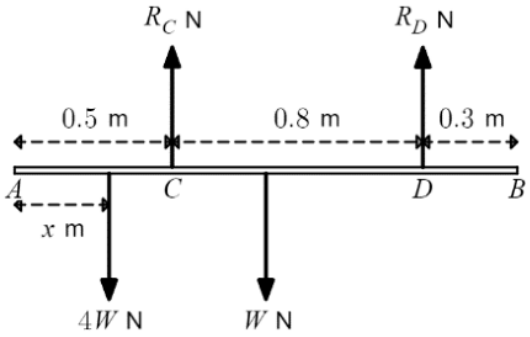
Point of tilting: $R_D = 0$ so $R(\uparrow)$: $R_C = 10m + 50$

M(C) $R(\circlearrowleft)$: $10m \times 0.5 - 50 \times 0.5 = 0$

$\Rightarrow m = 5$ kg

$\therefore R_C = 50 + 50 = 100$ N

26)



If it tips on C, then $R_D = 0$

$R(\uparrow)$: $R_C = 5W$

M(A) $R(\circlearrowleft)$: $-4Wx + 5W \times 0.5 - 0.8W = 0$

$\Rightarrow x = \frac{17}{40}$

If it tips on D, then $R_C = 0$

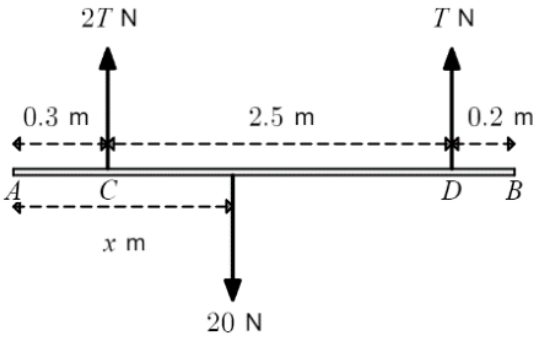
$R(\uparrow)$: $R_D = 5W$

M(A) $R(\circlearrowleft)$: $-4Wx - 0.8W + 5W \times 1.3 = 0$

$\Rightarrow x = \frac{57}{40}$

$\therefore \frac{17}{40} \leq x \leq \frac{57}{40}$

27 a)



$R(\uparrow)$: $2T + T = 2g \therefore T = \frac{2}{3}g$

M(A) $R(\circlearrowleft)$: $\frac{4}{3}g \times 0.3 - 2gx + \frac{2}{3}g \times 2.8 = 0$

$\frac{2}{5} - 2x + \frac{28}{15} = 0 \Rightarrow x = \frac{17}{15}$

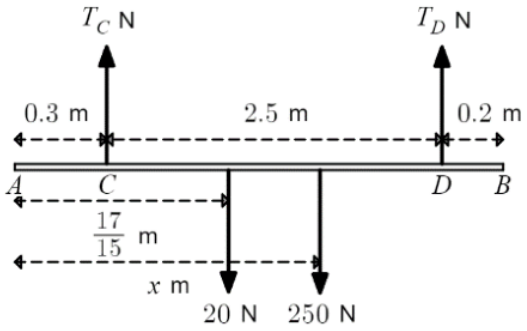
27 b)

The plank is modelled as a rod.

The ropes are modelled as light, inextensible string and are vertical.

27 c)

T_C and T_D may not exceed 200 N



Let the new distance be x metres.

M(T_D) $R(\circlearrowleft)$:

$-2.5 \times T_C + (2.8 - x) \times 250 + \left(2.8 - \frac{17}{15}\right) \times 20 = 0$

$T_C = -100x + \frac{880}{3}$

If $T_C \geq 200$,

$-100x + \frac{880}{3} \geq 200 \Rightarrow x \leq \frac{14}{15} = 0.93$ m

M(T_C) $R(\circlearrowleft)$:

$-20 \times \left(\frac{17}{15} - 0.3\right) - 250(x - 0.3) + 2.5 \times T_D = 0$

$T_D = 100x - \frac{70}{3}$

If $T_D \geq 200$,

$100x - \frac{70}{3} \geq 200 \Rightarrow x \geq \frac{67}{30} = 2.23$ m

The rope does not break for $\frac{14}{15} \leq x \leq \frac{67}{30}$

$\therefore$ they can walk from 0.93 m from A to 2.23 m from A